Lezione 2
Spice simulation

Stefan Cristi Zugravel
15/01/2026

What is SPICE?

- **S**imulation **P**rogram with **I**ntegrated **C**ircuit **E**mphasis
- Developed in 1973 by Laurence Nagel at UC Berkeley’s Electronics Research Laboratory
- Dependent on user defined device models

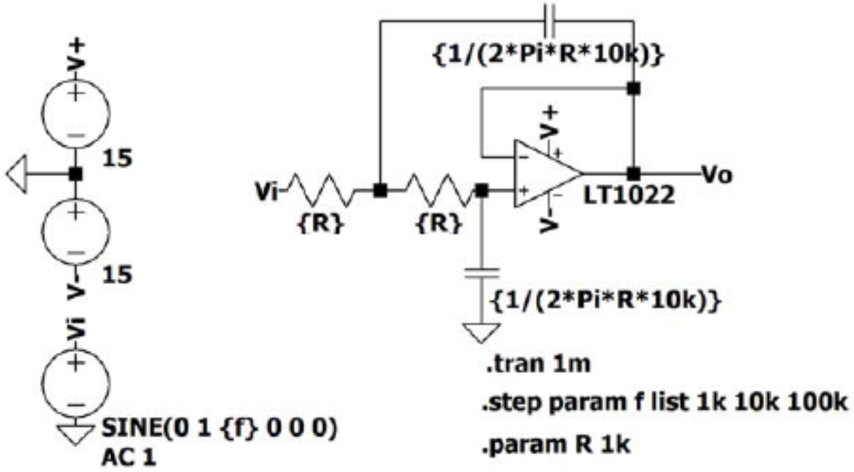
Components

Commands

```
* C:\Users\Devon\Desktop\6101\6101SallenKey.asc
XU1 N002 Vo V+ V- Vo LT1022
V1 V+ 0 15
V2 0 V- 15
R1 Vi N001 {R}
R2 N001 N002 {R}
C1 N002 0 {1/(2*Pi*R*10k)}
C2 N001 Vo {1/(2*Pi*R*10k)}
V3 Vi 0 SINE(0 1 {f} 0 0 0) AC 1
.step param f list 1k 10k 100k
.param R 1k
.tran 1m
.lib LTC.lib
.backanno
.end
```

LTspice

- Developed in 1998 at **Linear Technology Corporation**
- GUI, simulator, and schematic -> netlist for SPICE
- FREE and comes with tons of models




```
* C:\Users\Devon\Desktop\6101\6101SallenKey.asc
XU1 N002 Vo V+ V- Vo LT1022
V1 V+ 0 15
V2 0 V- 15
R1 Vi N001 {R}
R2 N001 N002 {R}
C1 N002 0 {1/(2*Pi*R*10k)}
C2 N001 Vo {1/(2*Pi*R*10k)}
V3 Vi 0 SINE(0 1 {f} 0 0 0) AC 1
.step param f list 1k 10k 100k
.param R 1k
.tran 1m
.lib LTC.lib
.backanno
.end
```

You do this

Ltspice makes this

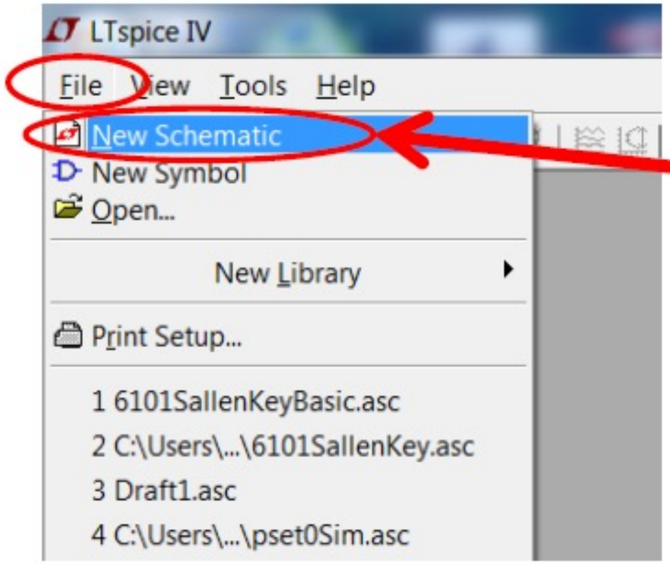
- **Linear Technology** was acquired by **Analog Devices** in 2017

What Should My Circuit Do?

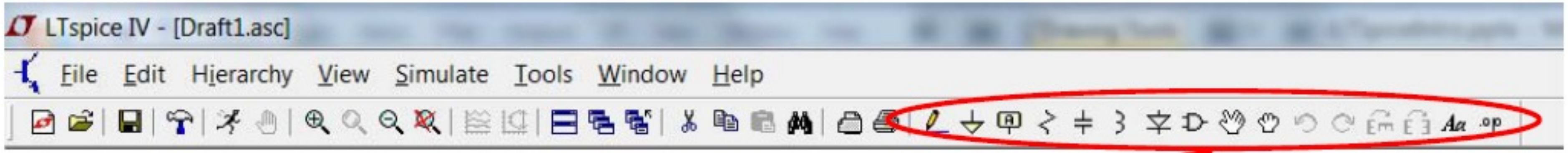
- The very first step to any simulation is to **know how your circuit should behave**.
- Simulation is a verification tool NOT A CIRCUIT SOLVER.

So how should this circuit behave?

Getting Started

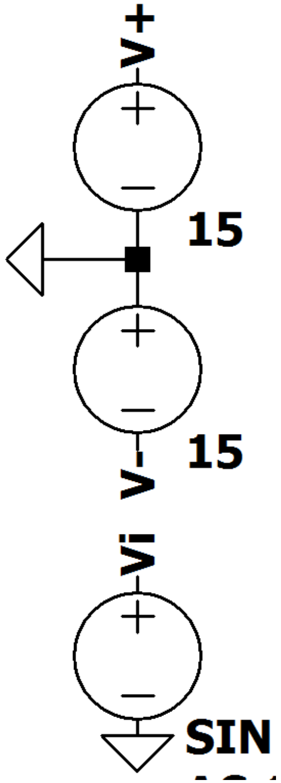


THAT'S IT!



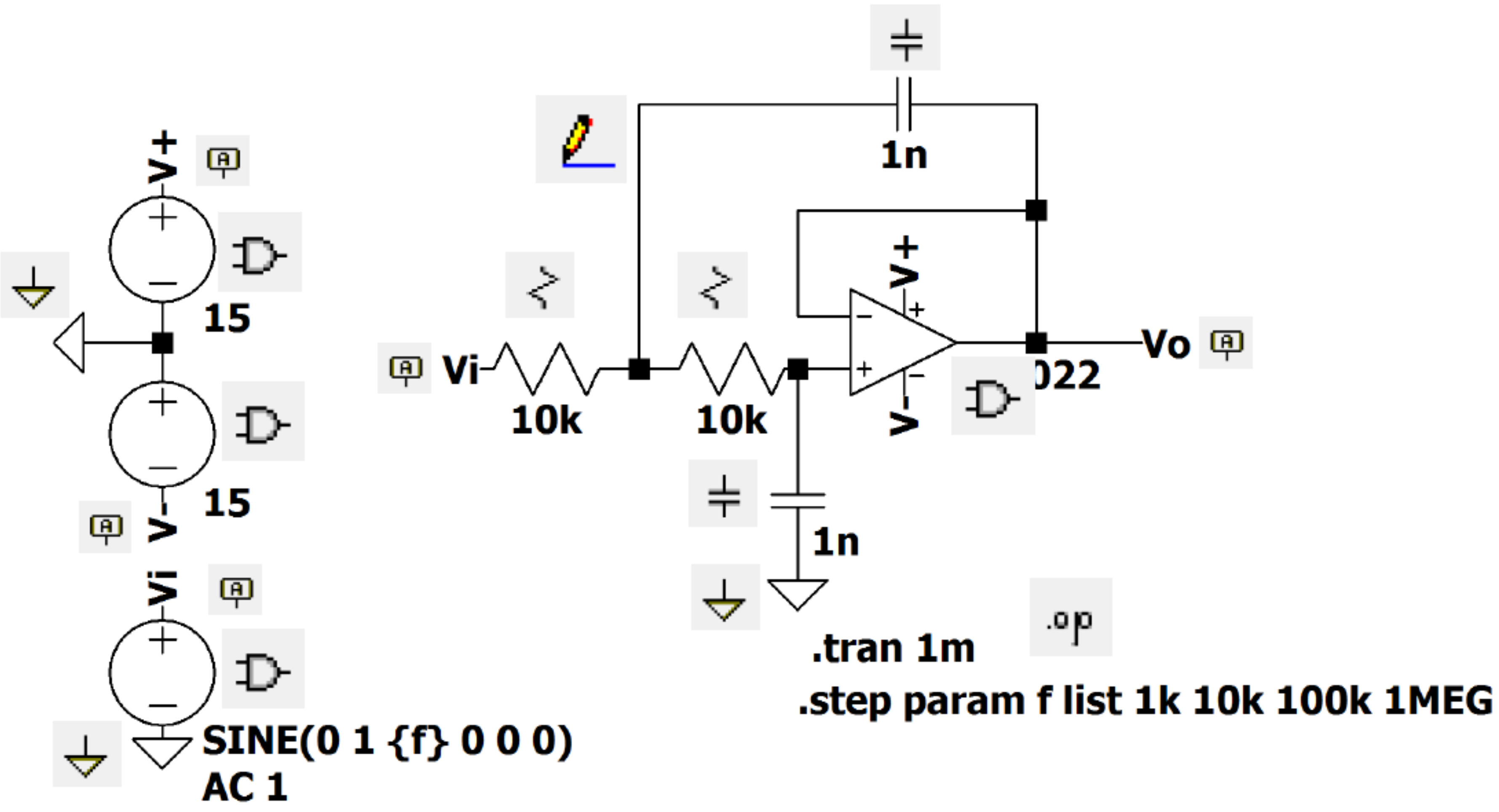
These buttons are where you will live

Net Labels



- By labeling nets you can avoid a giant mess of wires.
- Always use these for at least your power supplies.
- When you start making large circuits, your power supplies will provide energy all over your schematic.

Component to Menu Item Matchup

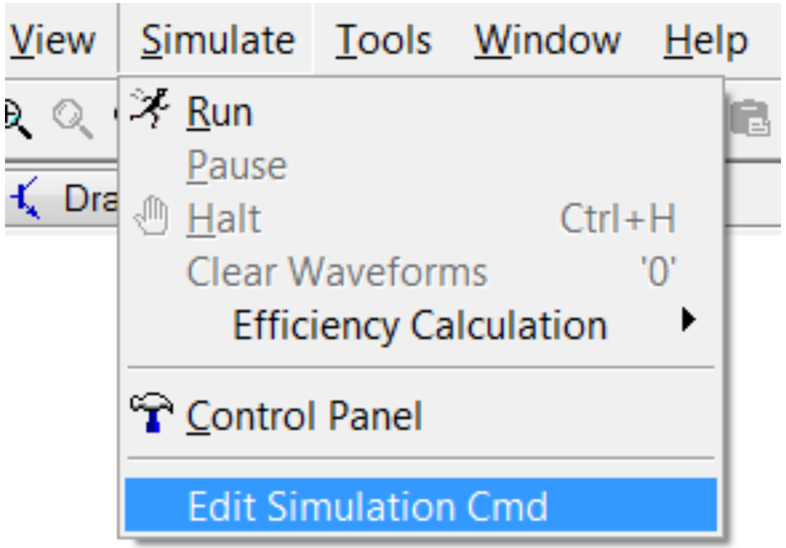


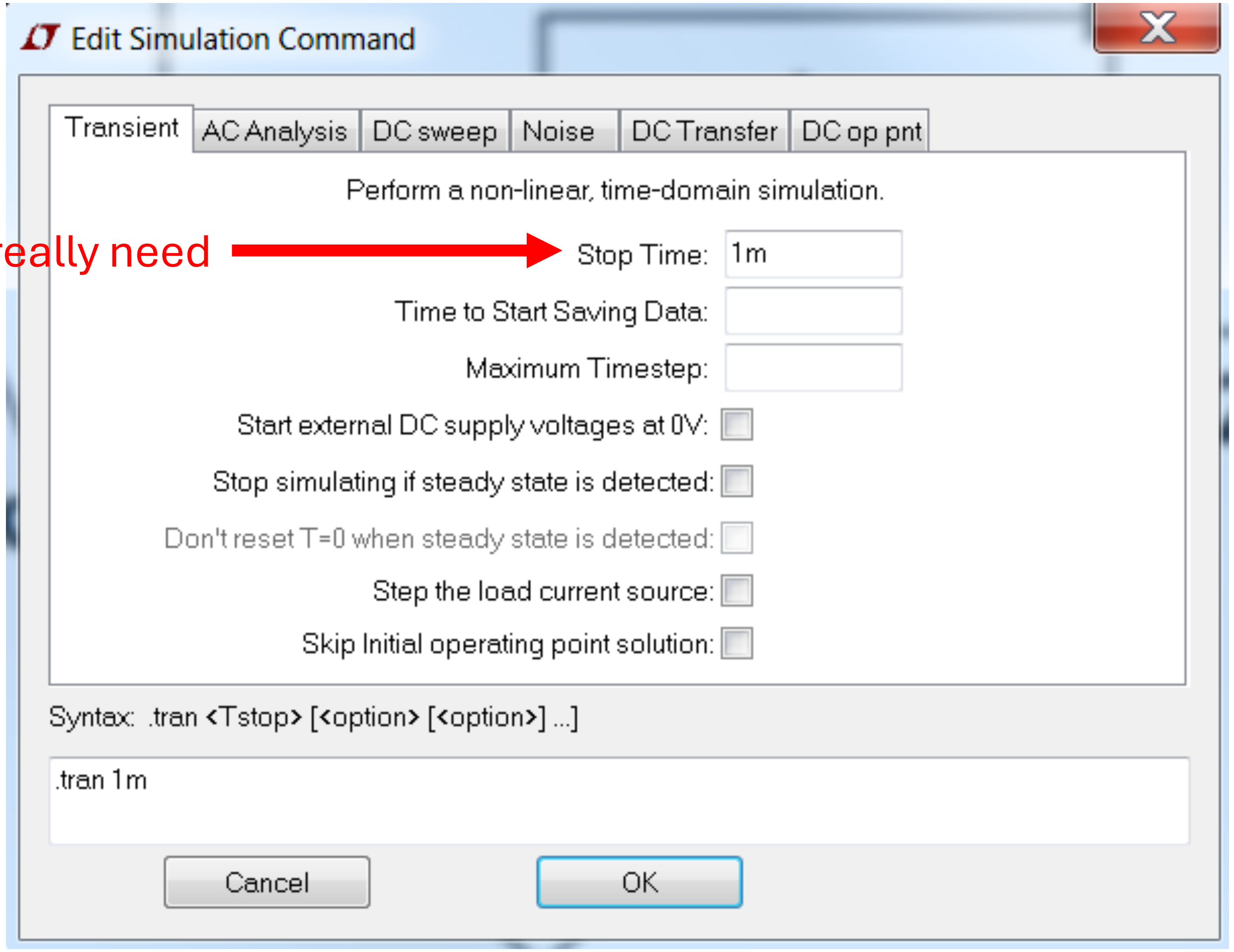
Simulation: DC Operating Point Analysis (.op)

“The DC operating point analysis calculates the **DC voltage** and current of each node in the **steady state** of the electronic circuit.”

Simulation: Transient (.tran)

This is all you really need



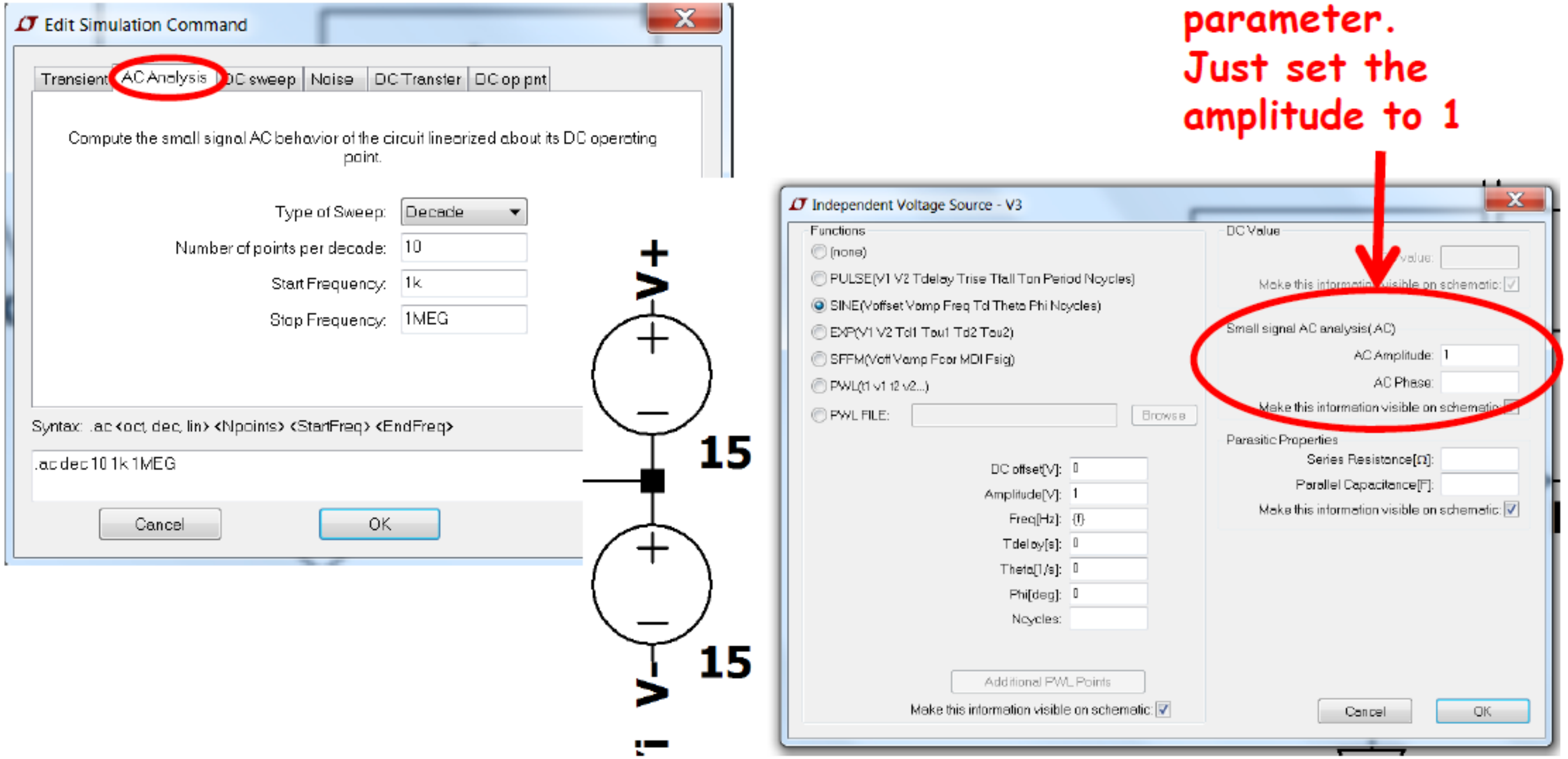


Simulation: AC (.ac) - 1

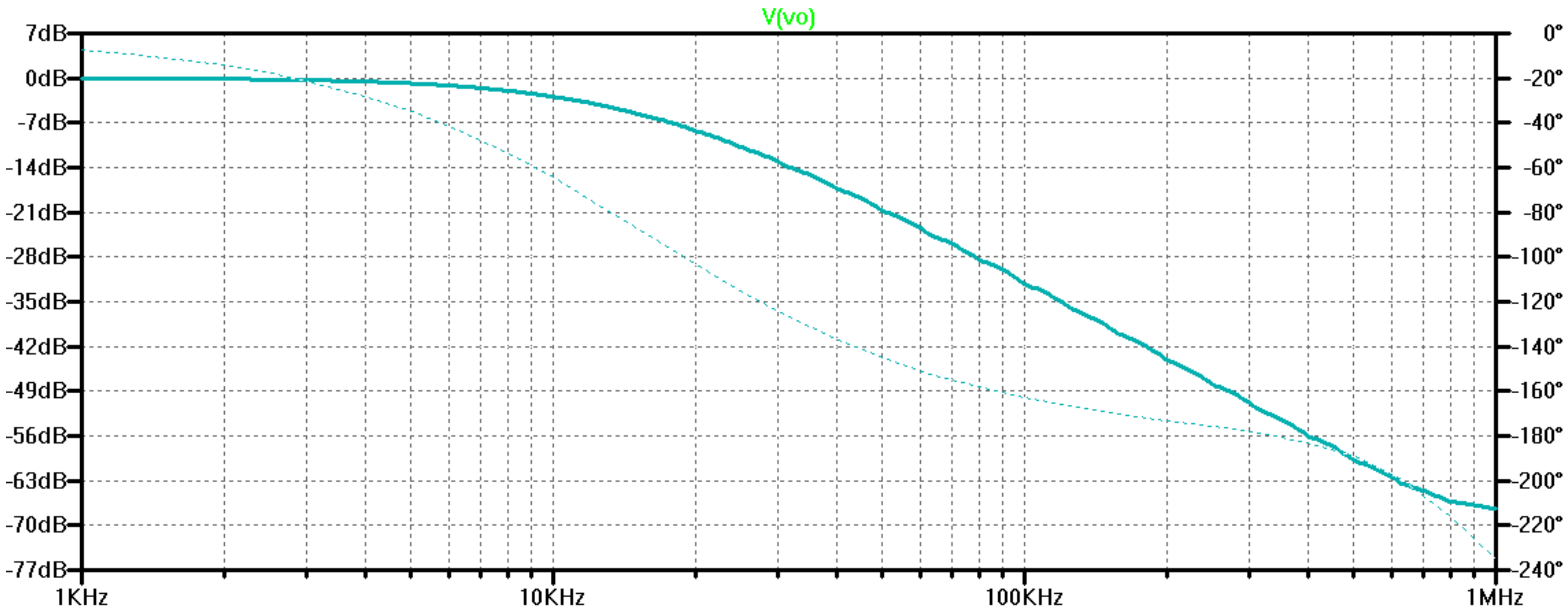
"The small signal analysis results are plotted in the waveform viewer as magnitude and phase over frequency."

AC simulation gives **Voltage and/or Current vs.frequency**.

This is the AC parameter. Just set the amplitude to 1



Simulation: AC (.ac) - 2



Simulation: AC (.ac) - 3

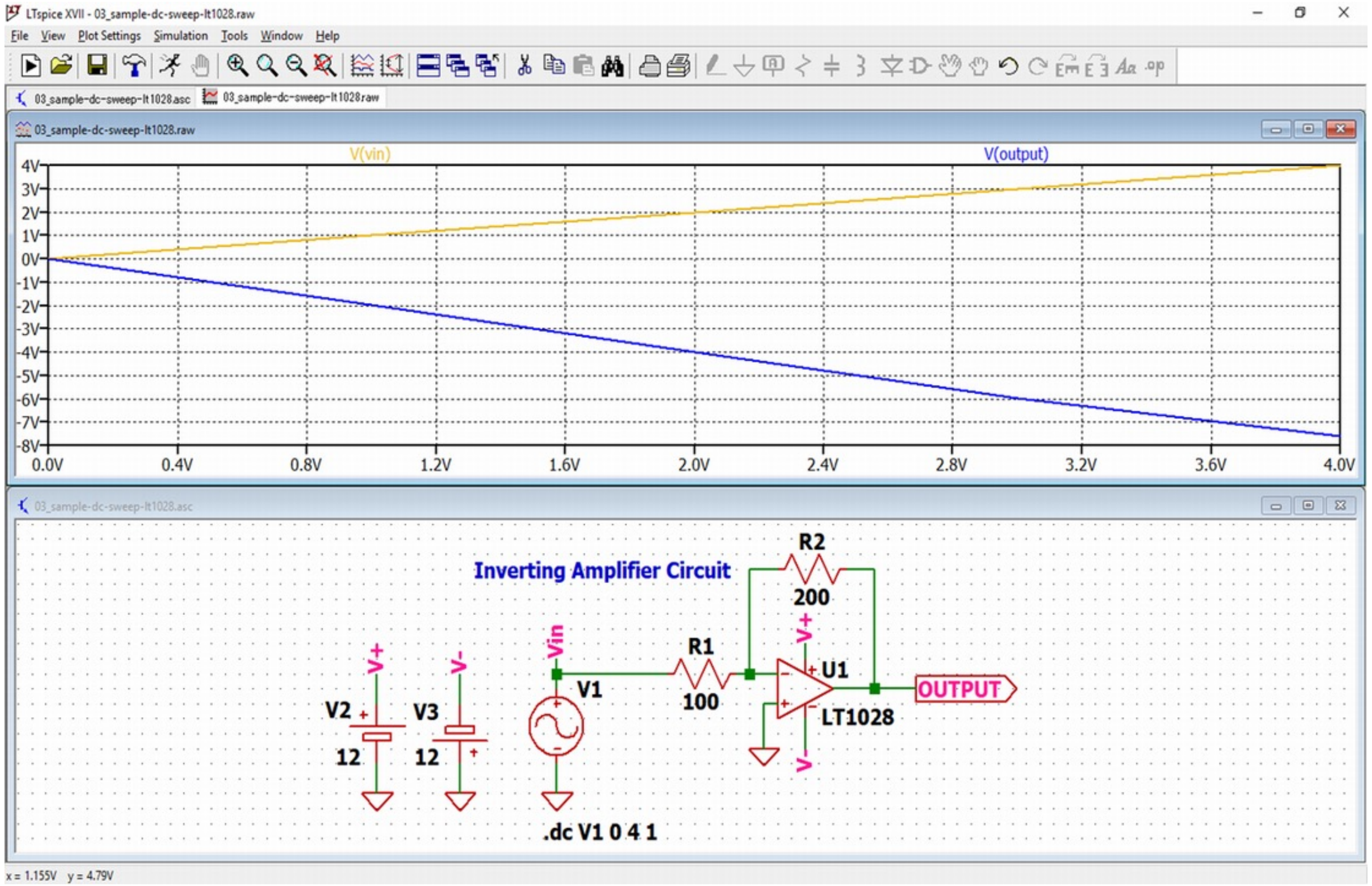
Small-signal AC analysis:

- Does **not** simulate time-domain waveforms
- Does **not** show distortion or clipping
- Does **not** handle large input swings
- Does **not** model switching behavior

When you set AC: 1 V it is not really 1 V, it is symbolic

Simulation: DC Sweep (.dc)

- DC sweep analysis sweeps the DC voltage of the input signal of the electronic circuit.
- It is used to analyze DC characteristics of diodes, transistors, op amps and so on.



LTspice HotKeys

	Schematic	Symbol	Waveform	Netlist
Modes	ESC – Exit Mode	ESC – Exit Mode		
	F3 – Draw Wire			
	F5 – Delete	F5 – Delete	F5 – Delete	
	F6 – Duplicate	F6 – Duplicate		
	F7 – Move	F7 – Move		
	F8 – Drag	F8 – Drag		
	F9 – Undo	F9 – Undo	F9 – Undo	F9 – Undo
	Shift+F9 – Redo	Shift+F9 – Redo	Shift+F9 – Redo	Shift+F9 – Redo
View	Ctrl+Z – Zoom Area	Ctrl+Z – Zoom Area	Ctrl+Z – Zoom Area	
	Ctrl+B – Zoom Back	Ctrl+B – Zoom Back	Ctrl+B – Zoom Back	
	Space – Zoom Fit		Ctrl+E – Zoom Extents	
	Ctrl+G – Toggle Grid		Ctrl+G – Toggle Grid	Ctrl+G – Goto Line #
	U – Mark Unncon. Pins	Ctrl+W – Attribute Window	'0' – Clear	
	A – Mark Text Anchors	Ctrl+A – Attribute Editor	Ctrl+A – Add Trace	
	Atl+Click – Power		Ctrl+Y – Vertical Autorange	Ctrl+R – Run Simulation
	Ctrl+Click – Attr. Edit		Ctrl+Click - Average	
	Ctrl+H – Halt Simulation		Ctrl+H – Halt Simulation	Ctrl+H – Halt Simulation
Place	R – Resistor	R – Rectangle		
	C – Capacitor	C – Circle		
	L – Inductor	L – Line		
	D – Diode	A – Arc		
	G – GND			
	S – Spice Directive			
	T – Text	T – Text		
	F2 – Component			
	F4 – Label Net			
	Ctrl+E – Mirror	Ctrl+E – Mirror		
	Ctrl+R – Rotate	Ctrl+R – Rotate		

Command Line Switches

Flag	Short Description
-ascii	Use ASCII .raw files. (Degrades performance!)
-b	Run in batch mode.
-big or -max	Start as a maximized window
-encrypt	Encrypt a model library
-FastAccess	Convert a binary .raw file to Fast Access Format
-netlist	Convert a schematic to a netlist
-nowine	Prevent use of WINE(Linux) workarounds
-PCBnetlist	Convert a schematic to a PCB netlist
-registry	Store user preferences in the registry
-Run	Start simulating the schematic on open
-SOI	Allow MOSFET's to have up to 7 nodes in subcircuit
-uninstall	Executes one step of the uninstallation process
-wine	Force use of WINE(Linux) workarounds

Simulator Directives – Dot Commands

Command	Short Description
.AC	Perform a Small Signal AC Analysis
.BACKANNO	Annotate Subcircuit Pin Names on Port Currents
.DC	Perform a DC Source Sweep Analysis
.END	End of Netlist
.ENDS	End of Subcircuit Definition
.FOUR	Compute a Fourier Component
.FUNC	User Defined Functions
.FERRET	Download a File Given the URL
.GLOBAL	Declare Global Nodes
.IC	Set Initial Conditions
.INCLUDE	Include another File
.LIB	Include a Library
.LOADBIAS	Load a Previously Solved DC Solution
.MEASURE	Evaluate User-Defined Electrical Quantities
.MODEL	Define a SPICE Model
.NET	Compute Network Parameters in a .AC Analysis
.NODESET	Supply Hints for Initial DC Solution
.NOISE	Perform a Noise Analysis
.OP	Find the DC Operating Point
.OPTIONS	Set Simulator Options
.PARAM	User-Defined Parameters
.SAVE	Limit the Quantity of Saved Data
.SAVEBIAS	Save Operating Point to Disk
.STEP	Parameter Sweeps
.SUBCKT	Define a Subcircuit
.TEMP	Temperature Sweeps
.TF	Find the DC Small-Signal Transfer Function
.TRAN	Do a Nonlinear Transient Analysis
.WAVE	Write Selected Nodes to a .WAV file

Suffix		Suffix	
		f	1e-15
T	1e12	p	1e-12
G	1e9	n	1e-9
Meg	1e6	u	1e-6
K	1e3	M	1e-3
		Mil	25.4e-6

Constants	
E	2.7182818284590452354
Pi	3.14159265358979323846
K	1.3806503e-23
Q	1.602176462e-19
TRUE	1
FALSE	0

analog.com

ANALOG DEVICES
AHEAD OF WHAT'S POSSIBLE™

LTspice